End Semester Examination - 2023

Course: B.Tech. | *Semester-IV* | *Time: 03 Hours*

Code: 101408 | *Subject: Structural Analysis* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

(iv) Question No. 1 is compulsory.

Q.1 *Choose the correct answer of the following (Any seven question only):* **[2 x 7 = 14]**

(a) The ordinates of influence line diagram for bending moment always have the dimensions of

(i) force (ii) length

(iii) force x length (iv) force / length

(b) The Castigliano's second theorem can be used to calculate deflections:

(i) at any point in a statically determinate structures only

(ii) at any point of any structure

(iii) at the point under the load in statically determinate structures only

(iv) at the point under the load of any structure

(c) Select the correct statement.

(i) Flexibility matrix is a square symmetrical matrix

(ii) Stiffness matrix is a square symmetrical matrix

(iii) Both (i) and (ii)

(iv) None of the above

(d) Muller Breslau's principle for obtaining influence lines is applicable to

(i) trusses

(ii) statically determinate beams and frames

(iii) statically indeterminate structures, the material of which is elastic and follows Hooke's law

(iv) any statically indeterminate structure

(e) What is the horizontal thrust in a symmetric parabolic two-hinged arch of span L and central rise (y) subjected to a uniformaly distributed load of intensity (w) per unit length over its entire span?

(i) $wL^2/(4y)$ (ii) $wL^2/(8y)$

(iii) $wL^2/(12y)$ (iv) $wL^2/(16y)$

(f) A single-bay portal frame of height h fixed at the base is subjected to a horizontal displacement Δ at the top. The base moment developed is proportional to

(i) $1/h$ (ii) $1/h^2$

(iii) $1/h^3$ (iv) h

(g) The number of simultaneous equations to be solved in the slope-deflection methods is equal to

(i) the degree of statical indeterminacy (ii) the degree of kinematic indeterminacy

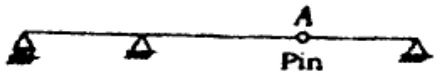
(iii) the number of joints in the structure (iv) None of the above

(h) Theorem of three moments is used for analysing

(i) fixed beam (ii) curved beam

(iii) continuous beam (iv) All of these

(i) What is the statical indeterminacy of the following beam?



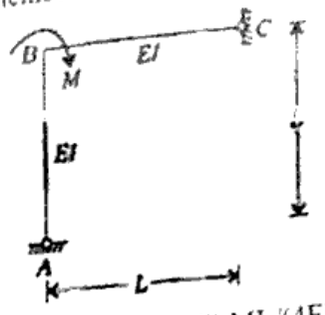
(i) 1 (ii) 3

(iii) 2 (iv) None of these

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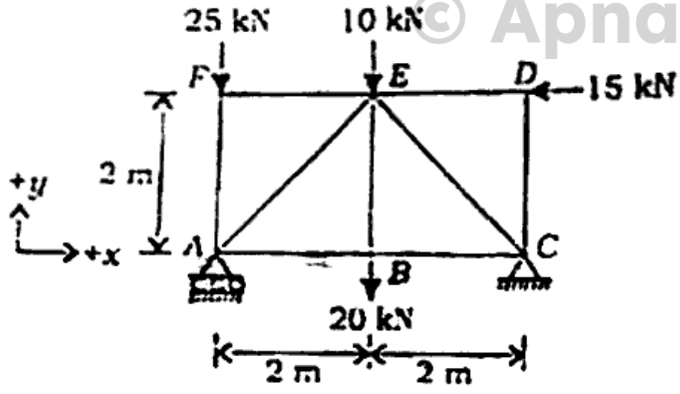
(j) What is the rotation of the member at C for a frame as shown in figure below?



(i) ML/(3EI)
(ii) ML/(4EI)
(iii) ML/(6EI)
(iv) ML/(12EI)

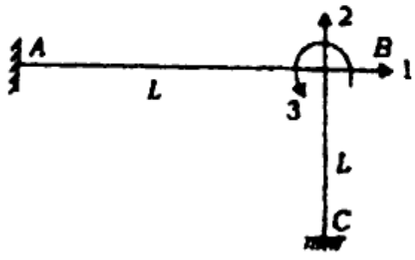
Q.2 A simply supported beam has a span of 12 m uniformly distributed load of 40 kN/m and 5 m long crosses the girder from left to right. Draw the influence line diagram for SF and BM at a section 4 m from left end. Use the diagram to calculate maximum SF and BM at this section. **[14]**

Q.3 Determine the vertical deflection of joint E of a truss shown in figure below. Take A = 2000 mm^2 and E = 200 kN/mm^2 **[14]**



Q.4 A two-hinged semicircular arch of radius R carries a distributed load uniformly varying from zero at left end to *w* per unit run at the right end. Determine the horizontal thrust at each support. **[14]**

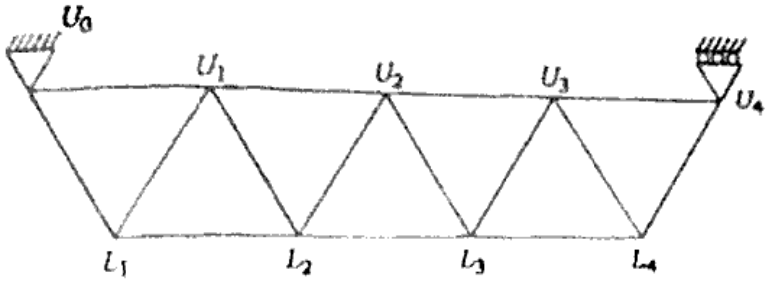
Q.5 Generate the stiffness matrix for the frame corresponding three degrees of freedom as shown in figure below. Take, AE and EI constant: **[14]**



Q.6 A beam of total length 2l is simply supported on three rigid supports. Two supports are at the end and one at the middle of the beam. Whereas the moment of inertia of the right span of the beam is I, that of the left span is 2I. The right span carries a point load *wl* at its mid-span and the left span carries a u.d.l. at the rate of *w* per unit length. Calculate the reactions at all supports. **[14]**

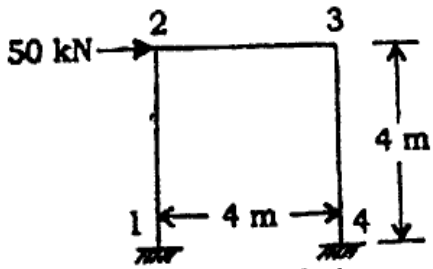
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Q.7 (a) Draw the influence line for the bar forces in members U_1U_2, L_1L_2 and U_1L_2 of the truss as shown in figure below. The length of each panel is 5 m and height of truss is 4 m. [7]



(b) What is the difference between real work done and virtual work done? Also derive the expression for strain energy due bending of beams. [7]

Q.8 Determine the support moments of the members of the frame shown in the following figure. EI is same for all the members. Use moment distribution method. [14]



Q.9 Write a short note on ***any four*** of the following: [3 ½ x4=14]

(a) Portal method of analysis
(b) Advantages of the matrix method over other methods of analysis of structure
(c) Betti's theorem of reciprocal work
(d) Eddy's theorem for an arch
(e) Distribution factor in moment distribution method with examples
(f) Assumptions and sign conventions in slope deflection method

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